

CSE 202 Final Part (b)

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I. MULTIPLE CHOICE QUESTIONS (40 POINTS)

1. What is the value of *sum* after the following code fragment is executed?

```
int sum = 0;
for (i = 1; i <= N; i++)
    for (j = 2 * i; j <= 2 * N; j += 2)
        sum++;
```

- a) N
- b) $(N^2 + N)/2$
- c) N^2
- d) N^3
- e) N^4

2. Given the time complexity equation for a recursive algorithm as $T(N) = 8T(N/4) + \mathcal{O}(N^2)$. What is the time complexity of that algorithm?

- a) $\mathcal{O}(N)$
- b) $\mathcal{O}(N \log N)$
- c) $\mathcal{O}(N^2)$
- d) $\mathcal{O}(N^2 \log N)$
- e) $\mathcal{O}(N^3)$

3. What is the time complexity of the insertMiddle for singly linked list, doubly linked list, and circular linked list?

- a) $\mathcal{O}(1)$, $\mathcal{O}(1)$, $\mathcal{O}(1)$
- b) $\mathcal{O}(N)$, $\mathcal{O}(N)$, $\mathcal{O}(N)$
- c) $\mathcal{O}(N)$, $\mathcal{O}(1)$, $\mathcal{O}(1)$
- d) $\mathcal{O}(N)$, $\mathcal{O}(1)$, $\mathcal{O}(N)$
- e) $\mathcal{O}(1)$, $\mathcal{O}(1)$, $\mathcal{O}(N)$

4. Given that there are more than one nodes in a single linked list, which of the following code fragment **inserts** a new node at the **beginning** of the list?

- a) `newNode.next = first;`
- b) `first = newNode;`
`first.next = newNode;`
- c) `first.next = newNode;`
`first = newNode;`
- d) `newNode.next = first;`
`first = newNode;`
- e) `first = newNode;`

5. *a* is an empty stack, which of the following code fragment prints the numbers 1 to *N* in increasing order?

- a)

```
for (int i = 1; i <= N; i++){
    a.push(i);
    System.out.println(a.pop());
}
```
- b)

```
for (int i = 1; i <= N; i++){
    a.push(i);
}
for (int i = 1; i <= N; i++){
    System.out.println(a.pop());
}
```
- c)

```
for (int i = 1; i <= N; i++){
    a.pop();
    System.out.println(a.push(i));
}
```
- d)

```
for (int i = 1; i <= N; i++){
    a.pop();
}
for (int i = 1; i <= N; i++){
    System.out.println(a.push(i));
}
```
- e) None of the above

6. Which of the following operations on which implementations can give queue overflow (adding an element when queue is full)?

- a) enqueue, array
- b) enqueue, linked list
- c) enqueue, array
enqueue, linked list
- d) enqueue, array
dequeue, array
- e) enqueue, linked list
dequeue, linked list

7. If the size of the queue is *N* (linked list implementation), how many elements can we store in that queue?

- a) $N - 2$
- b) $N - 1$
- c) N
- d) $N + 1$
- e) $N + 2$

8. For linked list implementation, what are the linked list operations to implement a FILO (first in, last out) queue?

- a) enqueue: insertLast, dequeue: removeFirst

- b) enqueue: removeLast, dequeue: insertFirst
 c) enqueue: removeFirst, dequeue: insertLast
 d) enqueue: insertFirst, dequeue: removeLast
 e) enqueue: insertFirst, dequeue: removeFirst
9. What are the minimum and maximum number of nodes in a binary tree of depth 4?
 a) 3, 8
 b) 5, 32
 c) 3, 7
 d) 4, 15
 e) 4, 16
10. If we want to obtain a rightly imbalanced binary search tree from numbers 1 to $2^n - 1$, what is the correct order of insertion of those numbers to the binary search tree?
 a) $2^n - 1, \dots, 2, 1$
 b) $1, 2, \dots, 2^n - 1$
 c) $2^{n-1}, 2^{n-2}, \dots, 1$
 d) $1, 3, 5, \dots, 2, 4, 6, \dots$
 e) $2^{n-1}, 2^{n-2}, 2^{n-1} + 2^{n-2}, \dots, 2^n - 1$
11. Which of the following numbers may be used as a hash table size?
 a) 14
 b) 32
 c) 117
 d) 128
 e) 211
12. Which of the following index(es) are better for queries like ($x > 10$)?
 a) Hash Index
 b) None
 c) Hash Index, sometimes Tree Index
 d) Tree Index
 e) Hash Index, Tree Index
13. What is the time complexity of update and insert for the min-heap?
 a) $\mathcal{O}(\log N), \mathcal{O}(\log N)$
 b) $\mathcal{O}(N), \mathcal{O}(\log N)$
 c) $\mathcal{O}(N), \mathcal{O}(1)$
 d) $\mathcal{O}(1), \mathcal{O}(1)$
 e) $\mathcal{O}(\log N), \mathcal{O}(1)$
14. What is the maximum number of elements that can be stored in depth h of a heap?
 a) $2^{h-1} - 1$
 b) 2^{h-1}
 c) $2^{h-1} + 1$
 d) $2^h - 1$
 e) $2^h + 1$
15. If the heights of two disjoint sets are 4 and 3, what is the height of the resulting set if those sets are merged?
 a) 5
 b) 4
 c) 6
 d) 7
 e) 3
16. What is the time complexity for finding the number of representative sets?
 a) $\mathcal{O}(\log N)$
 b) $\mathcal{O}(1)$
 c) $\mathcal{O}(N)$
 d) $\mathcal{O}(N \log N)$
 e) $\mathcal{O}(N^2)$
17. What is the maximum number of edges in a directed graph with loops for N nodes?
 a) N
 b) N^2
 c) $(N^2 + N)/2$
 d) $(N^2 - N)/2$
 e) N^3
18. What is the worst case time complexity of the algorithm (for a graph with N nodes) which checks if node i has no incoming edges (i) adjacency matrix, (ii) adjacency list representation?
 a) $\mathcal{O}(\log N), \mathcal{O}(\log N)$
 b) $\mathcal{O}(\log N), \mathcal{O}(1)$
 c) $\mathcal{O}(N), \mathcal{O}(1)$
 d) $\mathcal{O}(N), \mathcal{O}(N^2)$
 e) $\mathcal{O}(1), \mathcal{O}(N)$
19. If you want to sort the students in a university with respect to their sexes, what will be your choice of sorting algorithm?
 a) Bucket Sort
 b) Insertion Sort
 c) Merge Sort
 d) Quick Sort
 e) Bubble Sort
20. If you want to sort a set of 10.000.000 numbers, what will be your choice of sorting algorithm?
 a) Bucket Sort
 b) Insertion Sort
 c) Bubble Sort
 d) Quick Sort
 e) Selection Sort